

Pinned group summary: $\mathrm{SL}(\mathbf{n}=5)$

Pinning-Sympy automated output

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1 Basic group attributes

Group	SL (n=5)
Matrix size	5×5
Rank	4
Defining equations	<i>DefiningEquationPlaceholder</i>

1.1 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & x_{01} & x_{02} & x_{03} & & x_{04} \\ x_{10} & x_{11} & x_{12} & x_{13} & & x_{14} \\ x_{20} & x_{21} & x_{22} & x_{23} & & x_{24} \\ x_{30} & x_{31} & x_{32} & x_{33} & & x_{34} \\ x_{40} & x_{41} & x_{42} & x_{43} & -x_{00} - x_{11} - x_{22} - x_{33} & \end{bmatrix}$$

1.2 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 & 0 \\ 0 & 0 & t_2 & 0 & 0 \\ 0 & 0 & 0 & t_3 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{t_0 t_1 t_2 t_3} \end{bmatrix}$$

1.3 Trivial characters

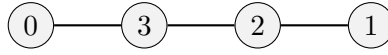
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	A4
Irreducible	True
Reduced	True
Simply laced	True
Number of roots	20

2.1 Dynkin diagram



2.2 Cartan matrix

$$\begin{bmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}$$

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
(-1, 0, 0, 0, 1)	(-1, 0, 0, 0, 1)	Yes	+	No
(-1, 0, 0, 1, 0)	(-1, 0, 0, 1, 0)	No	-	No
(-1, 0, 1, 0, 0)	(-1, 0, 1, 0, 0)	No	-	No
(-1, 1, 0, 0, 0)	(-1, 1, 0, 0, 0)	No	-	No
(0, -1, 0, 0, 1)	(0, -1, 0, 0, 1)	No	+	No
(0, -1, 0, 1, 0)	(0, -1, 0, 1, 0)	No	-	No
(0, -1, 1, 0, 0)	(0, -1, 1, 0, 0)	No	-	No
(0, 0, -1, 0, 1)	(0, 0, -1, 0, 1)	No	+	No
(0, 0, -1, 1, 0)	(0, 0, -1, 1, 0)	No	-	No
(0, 0, 0, -1, 1)	(0, 0, 0, -1, 1)	No	+	No
(0, 0, 0, 1, -1)	(0, 0, 0, 1, -1)	No	-	No
(0, 0, 1, -1, 0)	(0, 0, 1, -1, 0)	Yes	+	No
(0, 0, 1, 0, -1)	(0, 0, 1, 0, -1)	No	-	No
(0, 1, -1, 0, 0)	(0, 1, -1, 0, 0)	Yes	+	No
(0, 1, 0, -1, 0)	(0, 1, 0, -1, 0)	No	+	No
(0, 1, 0, 0, -1)	(0, 1, 0, 0, -1)	No	-	No
(1, -1, 0, 0, 0)	(1, -1, 0, 0, 0)	Yes	+	No
(1, 0, -1, 0, 0)	(1, 0, -1, 0, 0)	No	+	No
(1, 0, 0, -1, 0)	(1, 0, 0, -1, 0)	No	+	No
(1, 0, 0, 0, -1)	(1, 0, 0, 0, -1)	No	-	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$

α	β	Pairs (i, j)
$(-1, 0, 0, 0, 1)$	$(0, 0, 0, 1, -1)$	$(1, 1)$
$(-1, 0, 0, 0, 1)$	$(0, 0, 1, 0, -1)$	$(1, 1)$
$(-1, 0, 0, 0, 1)$	$(0, 1, 0, 0, -1)$	$(1, 1)$
$(-1, 0, 0, 0, 1)$	$(1, -1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 0, 0, 1)$	$(1, 0, -1, 0, 0)$	$(1, 1)$
$(-1, 0, 0, 0, 1)$	$(1, 0, 0, -1, 0)$	$(1, 1)$
$(-1, 0, 0, 1, 0)$	$(0, 0, 0, -1, 1)$	$(1, 1)$
$(-1, 0, 0, 1, 0)$	$(0, 0, 1, -1, 0)$	$(1, 1)$
$(-1, 0, 0, 1, 0)$	$(0, 1, 0, -1, 0)$	$(1, 1)$
$(-1, 0, 0, 1, 0)$	$(1, -1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 0, 1, 0)$	$(1, 0, -1, 0, 0)$	$(1, 1)$
$(-1, 0, 0, 1, 0)$	$(1, 0, 0, 0, -1)$	$(1, 1)$
$(-1, 0, 1, 0, 0)$	$(0, 0, -1, 0, 1)$	$(1, 1)$
$(-1, 0, 1, 0, 0)$	$(0, 0, -1, 1, 0)$	$(1, 1)$
$(-1, 0, 1, 0, 0)$	$(0, 1, -1, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0, 0)$	$(1, -1, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0, 0)$	$(1, 0, 0, -1, 0)$	$(1, 1)$
$(-1, 0, 1, 0, 0)$	$(1, 0, 0, 0, -1)$	$(1, 1)$
$(-1, 1, 0, 0, 0)$	$(0, -1, 0, 0, 1)$	$(1, 1)$
$(-1, 1, 0, 0, 0)$	$(0, -1, 0, 1, 0)$	$(1, 1)$
$(-1, 1, 0, 0, 0)$	$(0, -1, 1, 0, 0)$	$(1, 1)$
$(-1, 1, 0, 0, 0)$	$(1, 0, -1, 0, 0)$	$(1, 1)$
$(-1, 1, 0, 0, 0)$	$(1, 0, 0, -1, 0)$	$(1, 1)$
$(-1, 1, 0, 0, 0)$	$(1, 0, 0, 0, -1)$	$(1, 1)$
$(0, -1, 0, 0, 1)$	$(0, 0, 0, 1, -1)$	$(1, 1)$
$(0, -1, 0, 0, 1)$	$(0, 0, 1, 0, -1)$	$(1, 1)$
$(0, -1, 0, 0, 1)$	$(0, 1, -1, 0, 0)$	$(1, 1)$
$(0, -1, 0, 0, 1)$	$(0, 1, 0, -1, 0)$	$(1, 1)$
$(0, -1, 0, 0, 1)$	$(1, 0, 0, 0, -1)$	$(1, 1)$
$(0, -1, 0, 1, 0)$	$(0, 0, 0, -1, 1)$	$(1, 1)$
$(0, -1, 0, 1, 0)$	$(0, 0, 1, -1, 0)$	$(1, 1)$
$(0, -1, 0, 1, 0)$	$(0, 1, -1, 0, 0)$	$(1, 1)$
$(0, -1, 0, 1, 0)$	$(1, 0, 0, -1, 0)$	$(1, 1)$
$(0, -1, 1, 0, 0)$	$(0, 0, -1, 0, 1)$	$(1, 1)$
$(0, -1, 1, 0, 0)$	$(0, 0, -1, 1, 0)$	$(1, 1)$
$(0, -1, 1, 0, 0)$	$(0, 1, 0, -1, 0)$	$(1, 1)$
$(0, -1, 1, 0, 0)$	$(0, 1, 0, 0, -1)$	$(1, 1)$
$(0, -1, 1, 0, 0)$	$(1, 0, -1, 0, 0)$	$(1, 1)$
$(0, 0, -1, 0, 1)$	$(0, 0, 0, 1, -1)$	$(1, 1)$
$(0, 0, -1, 0, 1)$	$(0, 0, 1, -1, 0)$	$(1, 1)$
$(0, 0, -1, 0, 1)$	$(0, 1, 0, 0, -1)$	$(1, 1)$
$(0, 0, -1, 0, 1)$	$(1, 0, 0, 0, -1)$	$(1, 1)$
$(0, 0, -1, 1, 0)$	$(0, 0, 0, -1, 1)$	$(1, 1)$
$(0, 0, -1, 1, 0)$	$(0, 0, 1, 0, -1)$	$(1, 1)$
$(0, 0, -1, 1, 0)$	$(0, 1, 0, -1, 0)$	$(1, 1)$
$(0, 0, -1, 1, 0)$	$(1, 0, 0, -1, 0)$	$(1, 1)$
$(0, 0, 0, -1, 1)$	$(0, 0, 1, 0, -1)$	$(1, 1)$
$(0, 0, 0, -1, 1)$	$(0, 1, 0, 0, -1)$	$(1, 1)$
$(0, 0, 0, -1, 1)$	$(1, 0, 0, 0, -1)$	$(1, 1)$
$(0, 0, 0, 1, -1)$	$(0, 0, 1, -1, 0)$	$(1, 1)$
$(0, 0, 0, 1, -1)$	$(0, 1, 0, -1, 0)$	$(1, 1)$
$(0, 0, 0, 1, -1)$	$(1, 0, 0, -1, 0)$	$(1, 1)$

3 Root spaces

Root	Dimension
$(-1, 0, 0, 0, 1)$	1
$(-1, 0, 0, 1, 0)$	1
$(-1, 0, 1, 0, 0)$	1
$(-1, 1, 0, 0, 0)$	1
$(0, -1, 0, 0, 1)$	1
$(0, -1, 0, 1, 0)$	1
$(0, -1, 1, 0, 0)$	1
$(0, 0, -1, 0, 1)$	1
$(0, 0, -1, 1, 0)$	1
$(0, 0, 0, -1, 1)$	1
$(0, 0, 0, 1, -1)$	1
$(0, 0, 1, -1, 0)$	1
$(0, 0, 1, 0, -1)$	1
$(0, 1, -1, 0, 0)$	1
$(0, 1, 0, -1, 0)$	1
$(0, 1, 0, 0, -1)$	1
$(1, -1, 0, 0, 0)$	1
$(1, 0, -1, 0, 0)$	1
$(1, 0, 0, -1, 0)$	1
$(1, 0, 0, 0, -1)$	1

RootSpaceTablePlaceholder

4 Root subgroups

RootSubgroupTablePlaceholder

HomomorphismDefectTablePlaceholder

HomomorphismDefectIdentitiesPlaceholder

5 Commutators

CommutatorCoefficientTablePlaceholder

CommutatorIdentitiesPlaceholder

6 Weyl group

WeylGroupPlaceholder

WeylConjugationCoefficientPlaceholder

WeylConjugationIdentitiesPlaceholder

7 Coroot torus elements (h_α)

CorootTorusElementPlaceholder